

FINAL PROJECT REPORT

Online Education Analytics Dashboard

Student Performance, Engagement & Dropout Analysis

MySQL | SQL | Power BI | DAX

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1. Introduction

1.1 Project Overview

Online education platforms record large amounts of student data, such as clicks, scores, demographics and final results. This project analyses an online education dataset of **32,593 records** (14 columns) covering **26,073 unique students**, to understand how students perform, how engaged they are and why some drop out.

The data was stored and queried in MySQL, and the results are presented as an interactive Power BI dashboard and report with KPIs, filters and drill-down visuals.

1.2 Objectives

- Measure key indicators: total students, average score, average clicks, pass rate and dropout rate.
- Study how engagement level and performance level relate to scores, results and dropout.
- Compare students by region, gender and highest education.
- Identify students at High and Very High risk so that support can be given early.
- Deliver an easy-to-use dashboard that decision-makers can filter and explore.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

Online learning providers collect student data, but in raw tables it is hard to see who is engaged, who is struggling and who is likely to drop out. About one in four students (**26%**) drops out and only about half (**52%**) pass, yet there is no single view that shows how engagement, prior education, region and gender relate to these outcomes.

Solving this helps instructors spot at-risk students early, target support and improve course design.

2.2 Project Proposal (Proposed Solution)

Item	Details
Approach	Load the dataset into MySQL, analyse it with SQL queries, connect Power BI to the MySQL table, create KPI measures in DAX and build the dashboard and report pages.
Key features	Five KPI cards, six slicers, engagement, performance and risk analysis, dropout and pass rate by education, decomposition-tree drill-down and a Q&A visual.
Resources	Standard laptop (8 GB RAM), MySQL Workbench and Power BI Desktop.

2.3 Initial Project Planning

The work was planned as 15 user stories (31 story points) over four sprints.

Sprint	Epics / Work	User Stories	Story Points
Sprint-1	Data Loading, SQL Analysis (baseline KPIs and segments)	USN-1 to USN-4	7
Sprint-2	Dropout analysis in SQL, Data Connection and KPIs, Dashboard Overview with slicers	USN-5 to USN-9	11
Sprint-3	Engagement and Performance Analysis, Risk and Dropout Analysis, drill-down, Q&A	USN-10 to USN-13	9
Sprint-4	Validation and Reporting	USN-14, USN-15	4
Total		15 user stories	31

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

The online education dataset was received as a CSV file and imported into the MySQL table **online_education_dataset** (database **online_education_db**) using MySQL Workbench. The import took about 22 seconds. Power BI reads the table through the MySQL connector.

Item	Details
Source / format	Online education student dataset, CSV
Size	32,593 rows (records), 14 columns; 26,073 unique students
Storage	MySQL table online_education_dataset
Tools	MySQL Workbench (SQL), Power BI Desktop (dashboard, report and DAX measures)

Key columns used in the analysis (13 of the 14 columns):

Column	Description
id_student	Unique student ID
gender	Student gender (Male / Female)
region	Region of the student (13 regions)
highest_education	Highest education level (No Formal quals, Lower Than A Level, A Level or Equivalent, HE Qualification, Post Graduate Qualification)
studied_credits	Credits studied by the student
total_clicks	Total clicks on the learning platform (engagement measure)
avg_score	Average assessment score
final_result	Final outcome: Pass, Distinction, Fail or Withdrawn
engagement_level	Engagement group: High, Medium or Low
performance_level	Performance group: High, Medium or Low
risk_level	Risk group: Very High Risk, High Risk, Low Risk or Safe
pass_flag	1 if the student passed, otherwise 0
dropout_flag	1 if the student dropped out, otherwise 0

3.2 Data Quality Report

Check	Result
Data load	CSV imported into MySQL without errors (all Action Output steps returned OK).
Raw records	32,593 rows and 14 columns loaded.
Unique students	26,073 distinct id_student values. The 32,593 records belong to these 26,073 students, so some students have more than one record.
Dropout split in Power BI	7.64K dropout and 22.1K non-dropout records (dropout_flag), about 29.7K records in total.
Category values	Consistent labels: engagement and performance (High, Medium, Low), risk (4 levels), final result (4 values), flags (0 or 1).
Row-count check	Record counts compared before and after cleaning (total_old and total_clean queries in MySQL Workbench); unique students verified with COUNT(DISTINCT id_student).

How the counts are used. To avoid mixing records and students, each metric follows one rule throughout the SQL queries, DAX measures and visuals:

Metric	Counted on	Basis
Total students (KPI, risk and engagement splits)	Unique students	DISTINCT id_student (26,073)
Average score, average clicks	Records	AVG over rows
Dropout rate, pass rate	Records	Flag total divided by number of records

3.3 Data Exploration and Preprocessing

- **Preview:** SELECT query on the first 1,000 rows to check columns and values.
- **Cleaning check:** row counts compared before and after cleaning; unique students counted with COUNT(DISTINCT id_student).
- **SQL exploration:** 13 aggregate queries for totals, averages, pass and dropout counts, and breakdowns by engagement, performance, risk, click band, region and highest education (see Appendix 10.1).
- **Click bands:** total clicks grouped into 0-500, 500-1000, 1000-2000, 2000-3000 and 3000+ to compare average scores.
- **Power BI preparation:** connected the MySQL table and created five DAX measures (Total Students, Dropout Rate, Average Score, Average Clicks, Pass Rate); the formulas are listed in Appendix 10.1.

4. Data Visualization

4.1 Framing Business Questions

1. How many students are enrolled, and what are the average score, average clicks, pass rate and dropout rate?
2. How are students spread across engagement, performance and risk levels?
3. Do students with higher engagement (more clicks) score higher?
4. Which engagement and performance groups have the highest dropout?
5. How do region and gender affect scores and final results?
6. How does highest education influence dropout rate and pass rate?
7. What share of students is at High or Very High risk?
8. What are the final results (Pass, Distinction, Fail, Withdrawn)?

4.2 Developing Visualizations

The report has five pages and uses the visuals below.

Visualization	Chart Type	Purpose
KPI cards: Total Students, Dropout Rate, Average Score, Average Clicks, Pass Rate	Card	Headline metrics at a glance
Total Students by region; Average Score by region	Column chart	Compare enrolment and scores across 13 regions
Total Students by final_result	Donut / column chart	Share of Pass, Withdrawn, Fail and Distinction
Average and total clicks by engagement_level	Column chart	Link engagement to platform activity
Average Score by engagement_level and by performance_level	Column chart	Score gap between groups
Total Students by engagement_level	Pie chart	Engagement split
Dropout Rate and Pass Rate by highest_education	Column chart	Prior education versus outcomes
Total Students by risk_level	Pie chart	Risk distribution
Total Students by gender and final_result	100% stacked bar	Gender comparison of results
Dropout split; clicks by engagement and final result; score by region and final result	Pie / 100% stacked charts	Dropout share and outcome patterns
Decomposition trees (2)	Decomposition tree	Drill down by risk, education, gender, region, engagement and result
Slicers (6) and Q&A	Slicer / Q&A	Filtering and plain-language questions

5. Dashboard

5.1 Dashboard Design Screenshot with Insights

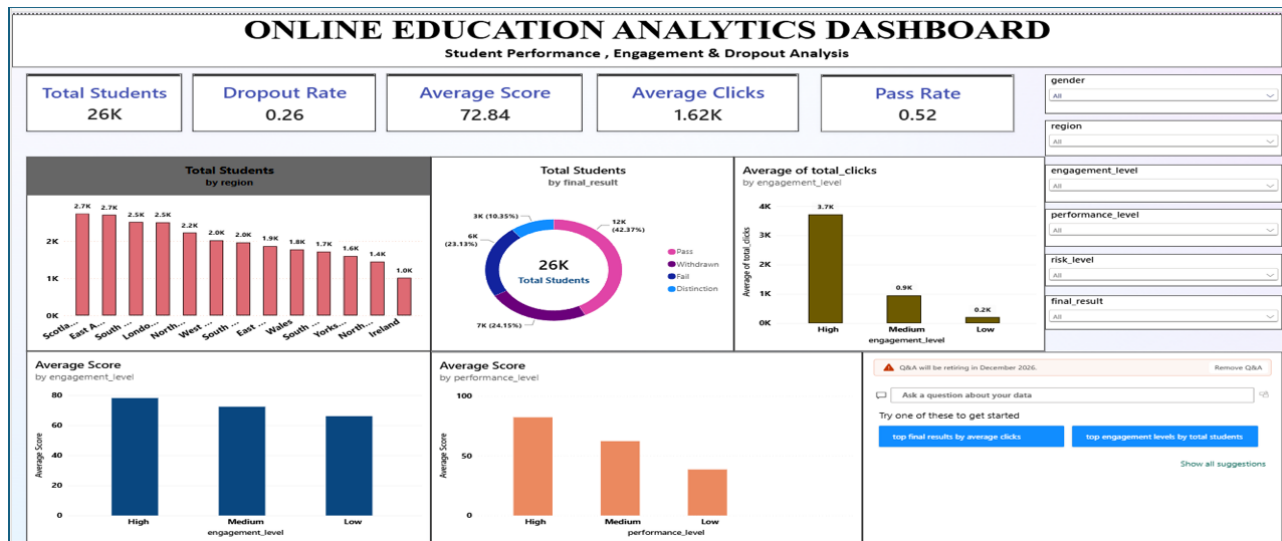


Figure 1: Online Education Analytics Dashboard

Insights:

- The dashboard covers about **26K students** with an average score of **72.84**, average clicks of **1.62K**, a pass rate of **0.52** and a dropout rate of **0.26**.
- Student numbers range from about 2.7K in the largest regions (Scotland, East Anglian) to about 1.0K in Ireland.
- Final results: **Pass 42.37%**, **Withdrawn 24.15%**, **Fail 23.13%**, **Distinction 10.35%**. Pass and Distinction together make up 52.72%, in line with the pass rate of 0.52.
- High** engagement students average about **3.7K clicks** against **0.2K** for Low, and average score rises from Low to High for both engagement and performance groups.
- Six slicers (gender, region, engagement_level, performance_level, risk_level and final_result) filter the visuals, and the Q&A visual answers questions in plain language.

6. Report

6.1 Report Design Screenshot with Insights

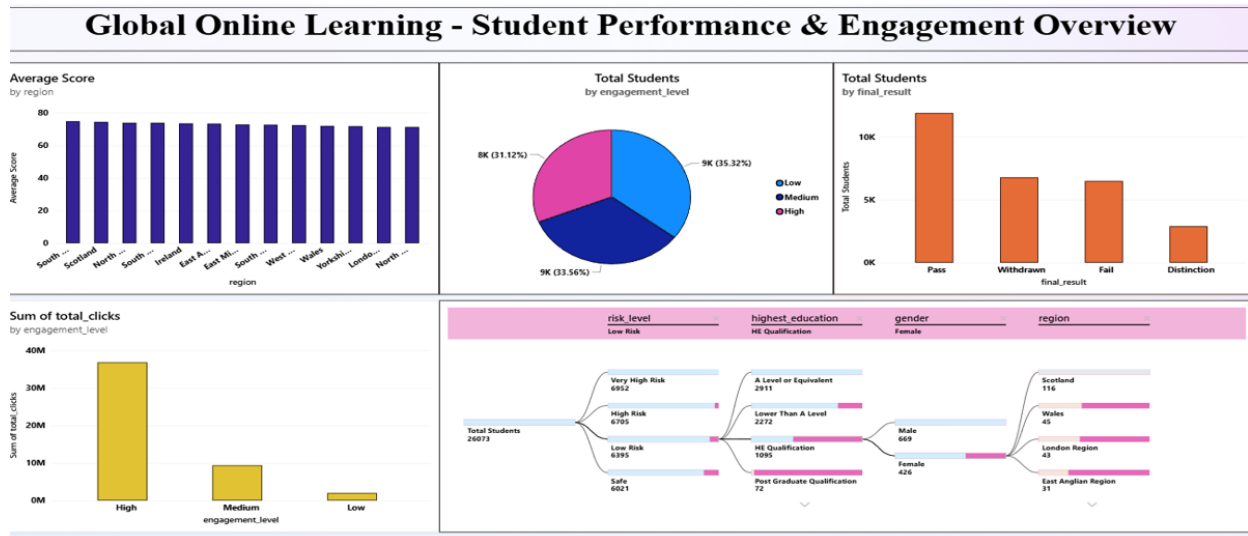


Figure 2: Student Performance & Engagement Overview page

Insights:

- Students are split almost evenly by engagement: Low **35.32%**, Medium **33.56%**, High **31.12%**.
- Average score is very similar across regions (roughly 71 to 75), so region affects enrolment more than results.
- High engagement students generate most of the total clicks.
- The decomposition tree splits the **26,073 students** by risk level: Very High Risk **6,952**, High Risk **6,705**, Low Risk **6,395**, Safe **6,021**.

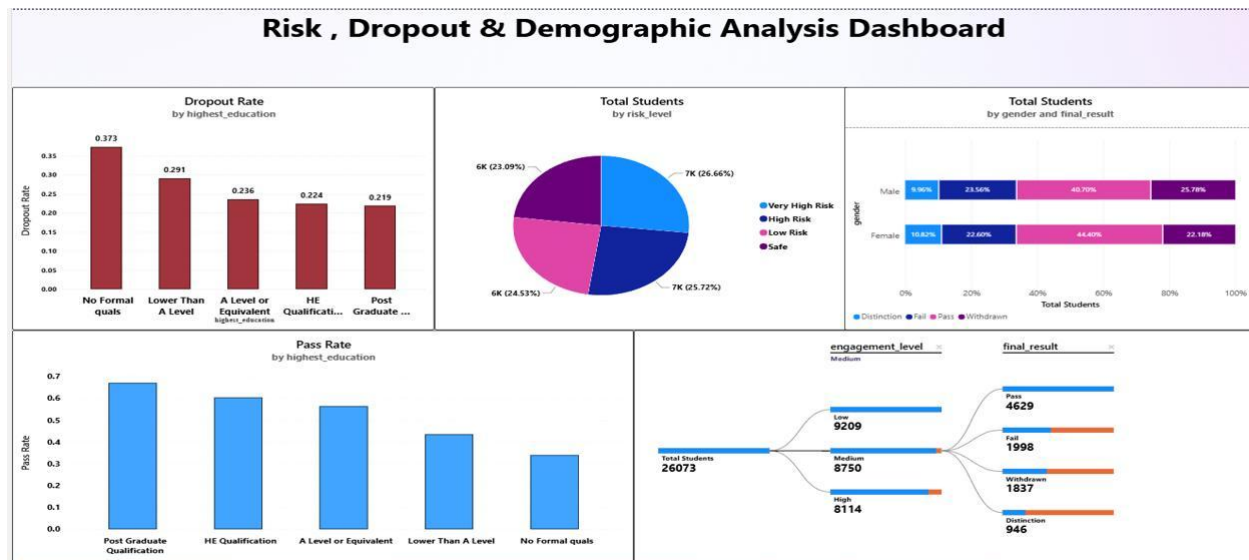


Figure 3: Risk, Dropout & Demographic Analysis page

Insights:

- Dropout rate falls as education rises: **0.373** (No Formal quals), 0.291 (Lower Than A Level), 0.236 (A Level or Equivalent), 0.224 (HE Qualification) and **0.219** (Post Graduate). Pass rate shows the opposite trend, from about 0.34 to about 0.67.
- Risk levels are spread evenly (Very High **26.66%**, High **25.72%**, Low **24.53%**, Safe **23.09%**), so **52.38%** of students are High or Very High risk.
- Female** students have a higher Pass share (**44.40%** vs **40.70%**) and a lower Withdrawn share (**22.18%** vs **25.78%**) than male students.
- Engagement level counts: Low **9,209**, Medium **8,750**, High **8,114** students.

7. Performance Testing

The table summarises the three performance checks made on the Power BI report; details follow in 7.1 to 7.3.

Test	Count	Details
7.1 Utilization of Data Filters	6 slicers	gender, region, engagement_level, performance_level, risk_level and final_result (dropdown slicers on the Dashboard page)
7.2 No. of Calculation Columns/Measures	5 measures	Total Students, Dropout Rate, Average Score, Average Clicks, Pass Rate. Engagement, performance and risk levels come directly from the dataset.
7.3 No. of Visualizations	40 visuals	34 charts, KPI cards, decomposition trees and Q&A, plus 6 slicers, across 5 report pages

7.1 Utilization of Data Filters

Six dropdown slicers on the Dashboard page let users filter the visuals by gender, region, engagement_level, performance_level, risk_level and final_result.

7.2 No. of Calculation Columns/Measures

Five DAX measures were created: Total Students, Dropout Rate, Average Score, Average Clicks and Pass Rate (formulas in Appendix 10.1). No calculated columns were needed, because the engagement, performance and risk levels are already present in the dataset.

7.3 No. of Visualizations

The report contains **40 visuals** across 5 pages: 34 charts, KPI cards, decomposition trees and the Q&A visual, plus 6 slicers.

8. Conclusion/Observation

- The platform has about **26K students**; **26%** drop out and about **52%** pass, with an average score of **72.84**.
- **Engagement matters:** High engagement students record far more clicks (about 3.7K vs 0.2K) and higher scores than Low engagement students.
- **Prior education matters:** dropout is highest for students with no formal qualifications (**0.373**) and lowest for postgraduates (**0.219**).
- **Region** changes how many students enrol but has little effect on average score.
- Over half (**52.38%**) of students are at High or Very High risk and should be the first group to receive support.
- **Female** students have a higher Pass share and a lower Withdrawn share than male students.
- **Recommendation:** monitor low-click students early, give extra support to students with lower prior education, and track the High and Very High risk groups.

9. Future Scope

- Build a machine-learning model to predict dropout and risk before results are known.
- Add time-based data (weekly clicks, assessment dates) to study trends and seasonality.
- Publish the report to Power BI Service with scheduled refresh and role-based access.
- Add course and assessment-level data to see which courses have the highest dropout.
- Set up automatic alerts for instructors when a student's engagement drops.

10. Appendix

10.1 Source Code

SQL queries (MySQL)

Two data-check queries (A and B) and the 13 analysis queries used in the project. All queries run on the database `online_education_db`.

```
USE online_education_db;
```

A. Data check: records and unique students

Confirms the raw row count and the number of distinct students (32,593 records, 26,073 unique students).

```
SELECT COUNT(*) AS total_records,  
       COUNT(DISTINCT id_student) AS unique_students  
FROM online_education_dataset;
```

B. Data check: preview

Preview of the first 1,000 rows to check columns and values.

```
SELECT *  
FROM online_education_dataset  
LIMIT 1000;
```

Query 1: Total students

Distinct students, so students with more than one record are counted once.

```
SELECT COUNT(DISTINCT id_student) AS total_students  
FROM online_education_dataset;
```

Query 2: Average student score

```
SELECT ROUND(AVG(avg_score), 2) AS average_student_score  
FROM online_education_dataset;
```

Query 3: Average total clicks

```
SELECT ROUND(AVG(total_clicks), 2) AS average_total_clicks  
FROM online_education_dataset;
```

Query 4: Records passed and dropped out

Counted at record level, using the `pass_flag` and `dropout_flag` columns.

```
SELECT COUNT(*) AS total_records,  
       SUM(CASE WHEN pass_flag = 1 THEN 1 ELSE 0 END) AS records_passed,  
       SUM(CASE WHEN dropout_flag = 1 THEN 1 ELSE 0 END) AS records_dropped_out  
FROM online_education_dataset;
```

Query 5: Students by engagement level

```
SELECT engagement_level,  
       COUNT(DISTINCT id_student) AS total_students  
FROM online_education_dataset  
GROUP BY engagement_level  
ORDER BY total_students DESC;
```

Query 6: Students and average score by performance level

```
SELECT performance_level,
       COUNT(DISTINCT id_student) AS total_students,
       ROUND(AVG(avg_score), 2) AS average_score
FROM online_education_dataset
GROUP BY performance_level
ORDER BY average_score DESC;
```

Query 7: Students by risk level

```
SELECT risk_level,
       COUNT(DISTINCT id_student) AS total_students
FROM online_education_dataset
GROUP BY risk_level
ORDER BY total_students DESC;
```

Query 8: Average score by engagement level

```
SELECT engagement_level,
       ROUND(AVG(avg_score), 2) AS average_score,
       COUNT(DISTINCT id_student) AS total_students
FROM online_education_dataset
GROUP BY engagement_level
ORDER BY average_score DESC;
```

Query 9: Average score by click band

Click bands are listed from lowest to highest so the trend is easy to read.

```
SELECT CASE
       WHEN total_clicks < 500 THEN '0 - 500'
       WHEN total_clicks < 1000 THEN '500 - 1000'
       WHEN total_clicks < 2000 THEN '1000 - 2000'
       WHEN total_clicks < 3000 THEN '2000 - 3000'
       ELSE '3000+'
     END AS click_band,
       COUNT(DISTINCT id_student) AS total_students,
       ROUND(AVG(avg_score), 2) AS average_score
FROM online_education_dataset
GROUP BY click_band
ORDER BY MIN(total_clicks);
```

Query 10: Dropout rate by engagement level

The dropout rate uses records for both the count and the total, so numerator and denominator are on the same basis.

```
SELECT engagement_level,
       COUNT(DISTINCT id_student) AS total_students,
       COUNT(*) AS total_records,
       SUM(CASE WHEN dropout_flag = 1 THEN 1 ELSE 0 END) AS dropout_records,
       ROUND(100.0 * SUM(CASE WHEN dropout_flag = 1 THEN 1 ELSE 0 END)
            / COUNT(*), 2) AS dropout_rate_pct
FROM online_education_dataset
GROUP BY engagement_level
ORDER BY dropout_rate_pct DESC;
```

Query 11: Dropout rate by performance level

```
SELECT performance_level,
       COUNT(DISTINCT id_student) AS total_students,
       COUNT(*) AS total_records,
       SUM(CASE WHEN dropout_flag = 1 THEN 1 ELSE 0 END) AS dropout_records,
       ROUND(100.0 * SUM(CASE WHEN dropout_flag = 1 THEN 1 ELSE 0 END)
            / COUNT(*), 2) AS dropout_rate_pct
FROM online_education_dataset
GROUP BY performance_level
ORDER BY dropout_rate_pct DESC;
```

Query 12: Students, score and clicks by region

```
SELECT region,  
       COUNT(DISTINCT id_student) AS total_students,  
       ROUND(AVG(avg_score), 2) AS average_score,  
       ROUND(AVG(total_clicks), 2) AS average_clicks  
FROM online_education_dataset  
GROUP BY region  
ORDER BY total_students DESC;
```

Query 13: Students, score and clicks by highest education

```
SELECT highest_education,  
       COUNT(DISTINCT id_student) AS total_students,  
       ROUND(AVG(avg_score), 2) AS average_score,  
       ROUND(AVG(total_clicks), 2) AS average_clicks  
FROM online_education_dataset  
GROUP BY highest_education  
ORDER BY average_score DESC;
```

DAX measures (Power BI)

The five measures below drive the KPI cards and visuals. They are created on the table `online_education_dataset`.

Measure 1: Total Students

Number of distinct students (KPI card, decomposition trees).

```
Total Students =
    DISTINCTCOUNT ( online_education_dataset[id_student] )
```

Measure 2: Dropout Rate

Share of records where `dropout_flag = 1`.

```
Dropout Rate =
    DIVIDE (
        SUM ( online_education_dataset[dropout_flag] ),
        COUNTROWS ( online_education_dataset )
    )
```

Measure 3: Average Score

Mean of the average assessment score.

```
Average Score =
    AVERAGE ( online_education_dataset[avg_score] )
```

Measure 4: Average Clicks

Mean of total clicks on the learning platform.

```
Average Clicks =
    AVERAGE ( online_education_dataset[total_clicks] )
```

Measure 5: Pass Rate

Share of records where `pass_flag = 1`.

```
Pass Rate =
    DIVIDE (
        SUM ( online_education_dataset[pass_flag] ),
        COUNTROWS ( online_education_dataset )
    )
```

10.2 GitHub Link

Item	Link
GitHub repository	[https://github.com/paricharlasaiteja014-wq/Online-Learning-Engagement-and-Student-Performance-Analysis]